NATIONAL INSTITUTE OF TECHNOLOGY PATNA

Department of Computer Science & Engineering

MID SEMESTER EXAMINATION, OCTOBER. 2024

B. Tech: CSE, Semester-V

Course Name: Information Security **Code: CS54119**

Maximum Time: 2 hours **Max. Marks: 30**

Instruction:

1. ***Attempt any Five questions.***
2. ***Assume any suitable data, if necessary.***
3. *The Marks, CO (Course Outcome) and BL (Bloom's Level) related to questions are mentioned on the right-hand margin.*

1.	a. Discuss three basic principles of information security goals in detail. b. Establish relation between security services and security techniques.	3+3	CO1, CO2	U, A
2.	a. Use a brute-force attack to decipher the following message. Assume that you know it is an affine cipher and that the plaintext "ab" is ciphered to "GL". XPALASXYFGFUKPXUSOGEUTKCDGFXANMGNVS b. Distinguish between keyless and keyed Transposition ciphers with examples.	3+3	CO2	E
3.	a. List some components of Modern block ciphers and explain it. b. Explain the DES function with neat diagram with key generation.	3+3	CO3	R, U
4.	a. Define a State in AES with example. b. List the parameters (block size, key size and number of rounds) for three AES versions	3+3	CO2	U, R
5.	Encrypt the message, "CRYPTOGRAPHY IS FUN" using the following ciphers: Ignore the space between words. a) Multiplicative cipher with key = 15 b) Affine cipher with key= (7, 2) c) Vigenere cipher with key "DOLLAR"	3×2	CO2	U,P
6.	Explain the following terms with example- a. Brute-force attack b. Statistical attack	3+3	CO1	U

****End of Questions****

NATIONAL INSTITUTE OF TECHNOLOGY PATNA
Department of Computer Science & Engineering
END SEMESTER EXAMINATION, DECEMBER. 2024
B. Tech: CSE, Semester-V

Course Name: Information Security **Code: CS54119**

Maximum Time: 3 hours **Max. Marks: 60**

Instruction:

1. ***Attempt Five questions.***
2. ***Assume any suitable data, if necessary.***
3. *The Marks, CO (Course Outcome) and BL (Bloom's Level) related to questions are mentioned on the right-hand margin.*

1	a. Explain how threats, vulnerabilities are involved in Information Security Risk Management (ISRM). b. Discuss basic principles of a stream cipher algorithm RC4 with key generation algorithm.	6+6	CO1 CO2	U, A
2	a. A potential attack on RSA cryptosystem is chosen-ciphertext attack. Explain how it can be successful in RSA. b. In RSA: i) Given n= 221 and e = 5, find d. ii) Given p = 19, q = 23, and e = 3, find n, $\phi(n)$, and d.	6+6	CO4	E
3	a. Find the results of the following using square and multiply method: 17^{22} mod 21 b. Consider the group G= $<Z_{10}^*, \times>$. Find out the primitive roots of the group. c. Find the result of 15^{-1}mod17, using Fermat's little theorem. d. Find the n bit word that is represented by the polynomial (x^2+1) in GF (2^5)	3×4	CO3	P, E
4	a. What are the essential ingredients of asymmetric key cryptography? Compare it with symmetric key cryptography. b. Consider a Super increasing series [7, 11, 23, 43, 87, 173, 357], a multiplier m = 41, and modulus n = 1001, encrypt input string "10101011110000010110011O" using Merkle–Hellman knapsack algorithm	6+6	CO4	U, P
5	a. Why Access control is important in computer security? Determine an entity authentication techniques used during client-server communication for protecting information system. b. Explain with diagram how Diffie Hellman key exchange protocol suffers with man-in-middle attack.	6+6	CO5	R, U
6	Explain the following terms- a. Single Sign-on b. Computer Ethics c. Polymorphic Virus d. Message Authentication Code	3×4	CO3 ,	U